

DEHRADUN, India

WMO: 421110

Lat: 30.317N

Lon: 78.033E

Elev: 682

StdP: 93.40

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	6.3	2.1	4.8	11.4	3.3	5.2	10.9	2.5	14.9	2.3	15.0	0.5	50	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.9	37.8	22.1	36.3	21.9	34.9	22.0	27.1	31.0	26.6	30.3	26.2	29.7	1.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	23.4	29.2	25.7	22.8	28.7	25.3	22.2	28.2	91.1	31.3	88.8	30.4	86.8	29.9	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
2.5	2.3	1.9	DB	3.7	39.9	1.1	1.5	2.9	41.0	2.3	41.9	1.6	42.7	0.8	43.8	(4)
			WB	3.2	28.4	1.1	0.9	2.3	29.0	1.7	29.6	1.0	30.1	0.2	30.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	13.2	15.9	20.6	25.4	28.3	27.1	26.7	25.9	22.7	18.5	14.7		
		DBStd	5.73	1.99	2.37	2.93	2.59	2.19	2.42	1.34	1.23	1.42	2.00	1.76	1.88	(7)
		HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0		
		HDD18.3	378	159	74	12	1	0	0	0	0	1	18	113		
		CDD10.0	4513	101	165	329	461	569	564	531	519	476	395	256	146	(10)
		CDD18.3	1847	1	6	83	211	311	314	272	261	226	137	25	1	(11)
		CDH23.3	18469	5	47	718	2328	3941	3896	2536	2187	1744	924	136	7	(12)
		CDH26.7	7714	0	4	214	1169	2196	2008	754	595	535	233	6	0	(13)
(14)	Wind	WSAvg	0.6	0.5	0.7	0.7	0.8	0.8	0.7	0.5	0.4	0.5	0.6	0.6	0.5	(14)
(15)	Precipitation	PrecAvg	2118	53	59	44	31	50	172	642	679	299	52	7	14	(15)
		PrecMax	3070	129	142	148	130	142	473	992	1027	866	236	42	117	(16)
		PrecMin	1333	6	0	0	0	0	170	388	79	13	2	0	0	(17)
		PrecStd	393	35	47	42	31	43	131	181	172	178	51	9	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	23.7	27.0	33.0	37.1	39.2	40.3	33.5	33.0	32.2	31.2	27.6	24.3		
		0.4% MCWB	14.6	16.8	19.0	20.6	21.4	22.5	26.7	26.4	25.7	22.9	18.8	16.2		
		2% DB	21.3	24.8	30.7	35.4	37.7	38.3	32.2	31.7	31.4	30.0	26.1	22.5		
		2% MCWB	13.5	15.9	18.9	20.8	21.4	23.0	26.2	26.3	25.2	22.3	18.0	15.0		
		5% DB	19.8	23.2	29.0	34.2	36.5	36.6	31.2	30.7	30.6	29.1	25.0	21.2		
		5% MCWB	12.8	15.2	18.3	20.4	21.5	23.0	26.1	26.0	24.9	21.8	17.6	14.3		
		10% DB	18.3	21.8	27.3	32.7	35.2	34.7	30.3	29.8	29.7	28.1	23.7	19.8		
		10% MCWB	12.3	14.4	17.6	19.7	21.5	23.6	25.9	25.7	24.5	21.2	17.0	13.7		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	16.0	18.1	21.6	24.7	26.3	27.5	27.7	27.6	27.2	25.1	20.5	17.2		
		0.4% MCDB	22.0	24.6	29.5	33.9	34.4	33.0	31.6	31.1	30.9	29.0	25.5	22.9		
		2% WB	14.4	16.7	20.2	22.8	24.9	26.6	27.1	26.9	26.2	23.9	19.3	15.9		
		2% MCDB	19.6	23.2	28.0	31.7	33.1	31.8	30.6	30.3	29.8	28.2	24.7	21.0		
		5% WB	13.5	15.8	19.0	21.5	24.0	26.0	26.6	26.4	25.7	23.0	18.4	14.9		
		5% MCDB	18.3	21.8	27.0	30.9	32.4	31.3	29.9	29.5	29.1	27.4	23.6	19.8		
		10% WB	12.7	14.9	18.1	20.4	23.0	25.5	26.2	26.0	25.2	22.2	17.4	14.0		
		10% MCDB	17.4	20.5	25.9	30.5	31.6	30.6	29.2	28.9	28.5	26.5	22.5	18.8		
(35)	Mean Daily Temperature Range	MDBR	11.8	11.7	12.9	13.7	13.1	9.9	6.0	6.1	7.8	11.3	12.2	12.4		
		5% DB MCDBR	13.6	13.5	14.2	15.1	14.6	13.5	7.7	7.7	9.1	11.7	12.9	13.6		
		5% DB MCWBR	6.6	6.2	5.5	5.9	5.5	4.9	3.5	3.6	4.4	5.7	6.3	6.8		
		5% WB MCDBR	12.2	12.3	13.1	13.6	12.1	8.8	6.8	6.7	7.9	9.5	12.1	12.6		
		5% WB MCWBR	6.4	6.0	5.9	6.3	5.8	4.1	3.4	3.5	4.2	4.9	6.6	6.7		
(40)	Clear-Sky Solar Irradiance	taub	0.403	0.429	0.440	0.500	0.582	0.656	0.648	0.559	0.493	0.474	0.422	0.397		
		taud	2.122	2.092	2.089	1.963	1.800	1.704	1.753	1.987	2.120	2.030	2.106	2.142		
		Ebn at Noon	804	822	843	805	741	685	688	747	785	765	775	787		
		Edh at Noon	130	147	157	185	219	240	228	178	151	154	131	122		
(44)	All-Sky Solar Radiation	RadAvg	3.34	4.22	5.56	6.52	6.97	6.07	4.66	4.54	5.01	5.11	4.17	3.46		
		RadStd	0.30	0.46	0.32	0.32	0.34	0.69	0.46	0.39	0.46	0.30	0.19	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	+1.13	N/A	+0.50	+0.43	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	+0.20	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page